

Advanced SQL Exercises for Online Retail Store

Exercise 1: Ranking and Window Functions

-- Database Schema

CREATE TABLE Products (

ProductID INT PRIMARY KEY,

ProductName VARCHAR(100),

Category VARCHAR(50),

Price DECIMAL(10, 2)

);

INSERT INTO Products (ProductID, ProductName, Category, Price) VALUES

(1, 'Laptop', 'Electronics', 1200.00),

(2, 'Smartphone', 'Electronics', 800.00),

(3, 'Tablet', 'Electronics', 600.00),

(4, 'Headphones', 'Accessories', 150.00);

```

SELECT TOP 3
ROW_NUMBER()
OVER(ORDER BY Price DESC),
Price ,
ProductName
FROM Products;

```

Results		Messages	
	(No column name)	Price	ProductName
1	1	1200.00	Laptop
2	2	800.00	Smartphone
3	3	600.00	Tablet

-- To handle ties condition we will insert some data

```

INSERT INTO Products
(ProductID, ProductName, Category, Price)
VALUES (5, 'Tablet', 'Electronics', 600.00);
SELECT * FROM Products;

```

Results

Messages

	ProductID	ProductName	Category	Price
1	1	Laptop	Electronics	1200.00
2	2	Smartphone	Electronics	800.00
3	3	Tablet	Electronics	600.00
4	4	Headphones	Accessories	150.00
5	5	Tablet	Electronics	600.00

```

SELECT RANK()
OVER(ORDER BY Price DESC) AS Ranked,
DENSE_RANK() OVER(ORDER BY Price DESC) AS Dense_Ranked
,Price ,
ProductName
FROM Products;

```

Results		Messages		
	Ranked	Dense_Ranked	Price	ProductName
1	1	1	1200.00	Laptop
2	2	2	800.00	Smartphone
3	3	3	600.00	Tablet
4	3	3	600.00	Tablet
5	5	4	150.00	Headphones

```

SELECT
    ProductName,
    Category,
    Price,
    RANK() OVER(
        PARTITION BY Category
        ORDER BY Price DESC
    ) AS PriceRank
FROM Products;

```

Results		Messages		
	ProductName	Category	Price	PriceRank
1	Headphones	Accessories	150.00	1
2	Laptop	Electronics	1200.00	1
3	Smartphone	Electronics	800.00	2
4	Tablet	Electronics	600.00	3
5	Tablet	Electronics	600.00	3